R26-DS-010 · FINAL YEAR RESEARCH PROJECT — PROGRESS PRESENTATION II

Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery

CareSense — a unified, web-based AI platform that detects caregiver stress, predicts patient falls, automates ADL documentation, and teaches sign language, evaluated as four independently rigorous modules.

MAIN SUPERVISOR	CO-SUPERVISOR	EXTERNAL SUPERVISOR
Prof. Samantha Thelijjagoda	Mr. Samadhi Rathnayake	Mr. Velu Sivakumar

SLIIT Faculty of Computing · Specialization: Data Science

Team Members



COMPONENT 01

Dayana Priyadharshani Kumar

IT22178640

AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution



COMPONENT 02

Harishalinee Elangovan

IT22057488

Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction



COMPONENT 03

Vidursha Prabagaran

IT22294098

Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs



COMPONENT 04

Kaushalya Nagenthraraja

IT22289384

Personalized Sign Language Tutoring System for Caregivers

PP1 Comments — How We Addressed Them

Dayana Priyadharshani Kumar IT22178640 · Stress-Risk Detection

Panel comment: *No comments recorded.*

How we addressed it: N/A — no PP1 panel feedback was received for this component.

Harishalinee Elangovan IT22057488 · Fall-Risk Detection

Panel comment: *Even though live captures were not taken, the uploaded video should be converted to skeleton and analysed (a prototype is not needed).*

How we addressed it: The fall-risk detection component now runs on real MediaPipe-extracted skeleton sequences (RGB is discarded immediately after pose extraction, never analysed as video). Subject-disjoint evaluation on 799 held-out sequences reaches 94.24% fusion accuracy, so the skeleton pipeline itself is validated without requiring a full end-to-end demo prototype.

Vidursha Prabagaran IT22294098 · Voice Documentation

Panel comment: *Improve and further validate the results.*

How we addressed it: Validated results through qualitative testing on unseen sentences and a real caregiver recording, surfacing a data-ordering bug, a parsing artifact, and a grammar tool corrupting medication names — all fixed. Improved results via systematic ablations, cutting ASR word error rate from 25.97% to 15.57% (~40% relative improvement), and selected final models on which one caught safety-critical cases, not just the highest benchmark score.

Kaushalya Nagenthraraja IT22289384 · Sign-Language Tutoring

Panel comment: *Overall the project is good; however, accuracy should be improved, and feedback from experts/field should be obtained if possible.*

How we addressed it: Recognition accuracy improved by comparing BiLSTM, SPOTER-style transformer, and TCN architectures with data augmentation and focal loss, selecting the best-performing TCN model (macro-F1 0.75). Field validation was carried out with CCHS Colombo to ground vocabulary and evaluation in real caregiver needs.

Validation: Field Insights

Before writing a single line of code, the team validated all four problems with three real care organisations.



Remark Elder Care

Discussed “Sign Vital” systems with co-founder Vidyantha Vasantha — confirming the need for stress detection via vocal harmonics.



CCHS Colombo

Director highlighted literacy barriers: staff struggle with manual logbooks and urgently need hands-free documentation.



Love & Elder Care

Director Hiran identified predictive fall analysis and logbook automation as the highest safety priorities.

Three organisations. Three independent confirmations that these four problems are real and unsolved.

Problem, Stakeholders & Measurable Pain Point

THE PROBLEM

Four connected burdens hit caregivers together: **occupational stress, undetected falls, manual ADL documentation, and no formal training** to communicate with deaf / hard-of-hearing patients.

Existing tools are single-purpose and reactive — each burden is handled in isolation, only after it has already occurred.

~1/3

of US registered nurses report burnout; replacing one costs ≈$61,110.

14M+

US adults 65+ fall yearly — 41,400 deaths (2023), ≈$80B in costs.

25–40%

of a 12-hour shift spent documenting instead of caregiving.

6.3%

of surveyed staff had sign-language training, though 69.3% called it vital.

STAKEHOLDERS

Role	System Access	Outcome
Caregiver	Personal stress dashboard, fall-risk alerts, voice documentation, sign-language lessons	Reduced burnout, safer patients, faster documentation, learns to sign
Senior Caregiver / Supervisor	Team-wide dashboards, escalation alerts, workload-redistribution suggestions	Oversight across all caregivers, earlier intervention
Admin	Creates & manages caregiver / supervisor accounts (email & password), system configuration	Secure onboarding and access control

Research Gap & Objectives

RESEARCH GAP	OBJECTIVE
Stress Detection Existing tools ignore relational structure & causal, pre-outcome features.	**Objective 1** Model personalized, causally-ordered stress risk (LOSO-validated).
Fall Detection Lab-only validation — domain shift to real deployment cameras untested.	**Objective 2** Detect fall risk at the edge under subject-disjoint & cross-dataset evaluation.
Documentation Rigid prompted dialogue or unstructured transcripts, not free narration.	**Objective 3** Automate safety-aware documentation with high alert recall, GPU-free.
Sign-Language Tutoring Recognition-only — no per-landmark error localization or mastery adaptation.	**Objective 4** Evaluate signing execution quality and adapt lessons to learner errors.

Objective 5 — Integrate & Evaluate Holistically: Unify all four components into one explainable, caregiver-facing dashboard and evaluate the system as a whole — reporting demonstrated results and limitations transparently.

Solution Overview & Architecture

AI-Assisted Caregiver Workforce Deterioration Detection

Physio + workload signals → Personalized baseline → LightGBM risk model → SHAP explanation

Skeletal Movement-Based Fall-Risk Detection & Prediction

RGB camera → MediaPipe skeleton → ST-GCN + biomechanics → Fusion MLP risk verdict

Automated ADL Extraction & Voice-to-Documentation

Caregiver speech → ASR + cleanup → ADL field extraction → Alerts + report

Personalized Sign-Language Tutoring System

Signing video → Landmark tracking → Recognition + DTW score → Adaptive feedback

Unified Web-Based Caregiver Dashboard

All four pipelines are trained and evaluated independently, then converge into one explainable, real-time view for caregivers and supervisors.

Integration & Completion Status

Deterioration Detection
✓ Core complete

Fall-Risk Detection
✓ Core complete

Voice Documentation
✓ Core complete

Sign-Language Tutoring
✓ Core complete

Unified Backend API + Caregiver Dashboard

FastAPI inference services route every module's output into one explainable, real-time dashboard.

Structured Data
Supabase (PostgreSQL)

Caregiver & patient records, risk scores, ADL fields, accounts, session logs

Unstructured Data
Cloudflare (R2 Object Storage)

Raw skeleton sequences, audio clips, signing video, trained model artifacts

Evaluation Evidence & Non-Functional Requirements

EVALUATION EVIDENCE (measured on held-out / subject-disjoint test data)



NON-FUNCTIONAL REQUIREMENTS

Privacy

RGB discarded immediately after pose extraction; only de-identified data stored.

Real-Time

CPU-only edge inference (6.22 ms/window) meets the 10-windows/s throughput requirement.

Explainability

SHAP values, per-feature z-scores, and per-landmark DTW deviation explain every flag.

Robustness

Subject-disjoint & cross-dataset protocols avoid identity or domain leakage.

Commercialization & Demo Roadmap

SaaS Subscription

Modular, per-component licensing for hospitals & eldercare facilities.

Target Market

Sri Lanka & South Asia's growing elderly-care sector; global market >USD 180B by 2030.

Defensibility

Only platform integrating all four caregiver pain points with edge-first privacy.

DEMO ROADMAP TO FINAL

Stress-Risk Detection (1)
Ship the real-time FastAPI inference backend

Fall-Risk Detection (2)
Extend the webcam domain-adaptation dataset

Voice Documentation (3)
Add ClinicalBERT alerts + gTTS spoken handover

Sign-Language Tutoring (4)
Expert / clinical feedback to raise sign accuracy

Final Demo (5)
One unified dashboard, all four modules live

System Overview — Data Flow & Beneficiaries

DATA SOURCES

EMR

Voice Mics

Cameras (skeleton)

Caregiver Strain Detection & Adaptive Redistribution

Monitors biometrics & logs; prevents burnout, suggests fair shift adjustments.

STABLE

Edge Fall Prediction & Risk Assessment

Analyzes anonymized skeleton movement at the edge; detects instability, sends real-time alerts.

STABLE

Context-Aware ADL Documentation

Transcribes caregiver voice into structured, accurate ADL records; auto-generates handover summaries.

STABLE

Personalized Sign-Language Tutoring

Uses computer vision for real-time sign recognition; delivers customized lessons and feedback.

STABLE

INTEGRATED AI CAREGIVING PLATFORM

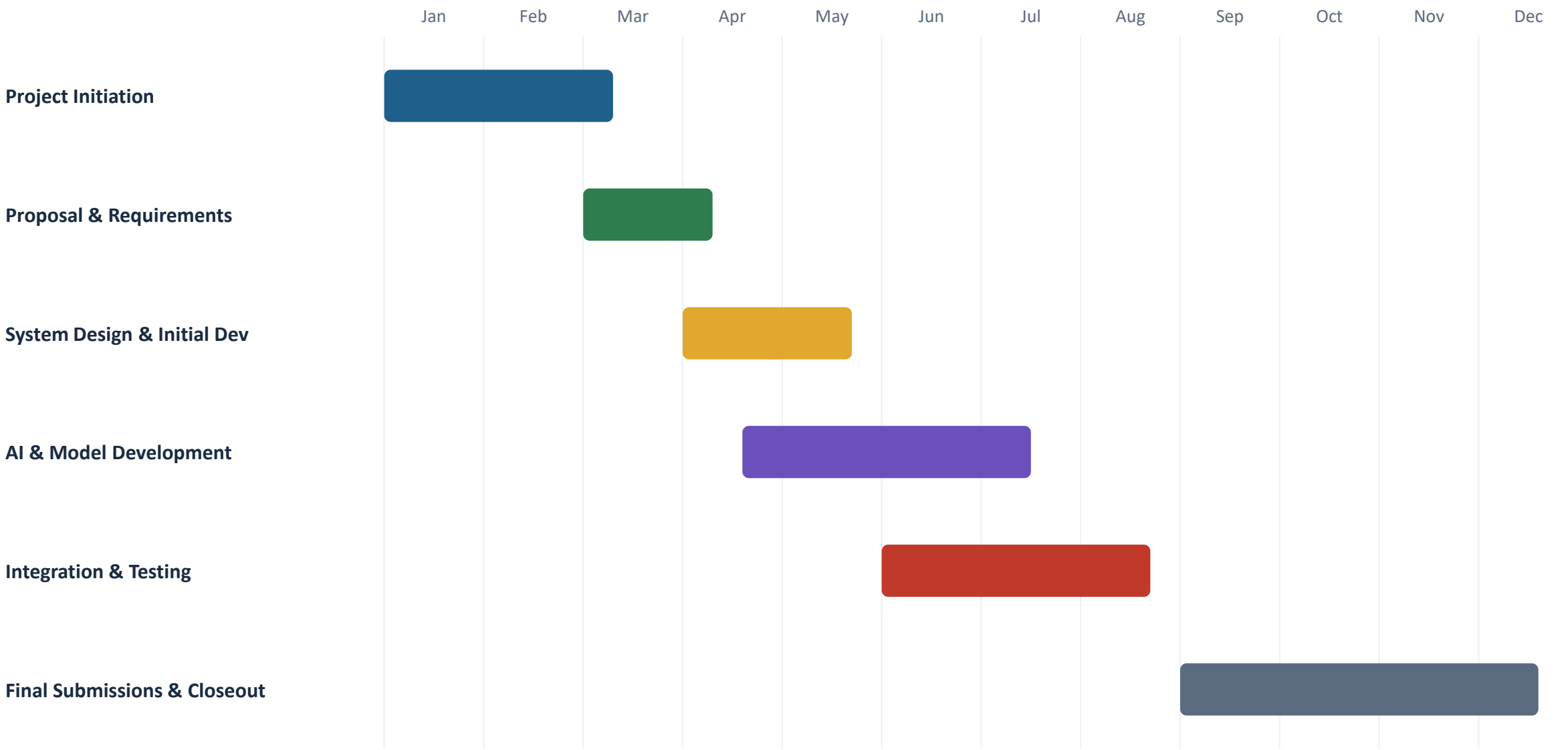
Beneficiaries

Caregivers · Patients · Supervisors · Institutions

Work Breakdown Structure

Project Management	IT22178640 — Workforce AI	IT22057488 — Fall Prediction	IT22294098 — Voice Documentation	IT22289384 — Sign-Language Tutoring	Integration & Testing	Final Presentation & Report
• Project planning	• Deterioration risk detection	• Skeletal pose extraction	• ADL speech-to-text pipeline	• Sign language collection	• Module integration	• Final report writing
• Requirement analysis	• Schedule stability analysis	• Fall risk prediction model	• ADL content extraction	• 3D avatar instruction module	• System validation	• Project presentation
• Shared data strategy	• Fair workload redistribution	• Risk replay visualization	• Voice-log documentation	• Adaptive feedback system	• User testing	• Commercialization planning
• Workload management	• Explainable case narratives	• False-alarm minimization	• Interactive report summaries	• Usability & learning tests		

Research Project Timeline — Gantt Chart



Q & A

Questions from the panel about CareSense — architecture, evaluation, and next steps.

Thank You

R26-DS-010 · CareSense · SLIIT Faculty of Computing